



University
of Exeter

COMM113

Deep Learning – Week 11

Trustworthy Deep Learning

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Fairness

Models inherit patterns from data. If data embeds historical inequities, neural networks amplify them.

Unfair Examples:

- COMPAS recidivism case: higher false-positive rates for Black defendants.
- Amazon hiring model: penalised CVs containing “women’s”.
- Image classifiers (e.g., classifying doctors as men, nurses as women).

What are protected classes?

- Race
- Sex
- Religion
- National origin
- Citizenship
- Pregnancy
- Disability status
- Genetic information
- ...

Why is it insufficient to simply avoid using protected attributes?

What are protected classes?

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Why is it insufficient to simply avoid using protected attributes?

Because other attributes can act as proxies (postcode, education, language patterns).

Fairness through unawareness

- **Idea:** Don't record protected attributes, and don't use them in your algorithm
- Predict risk Y from features X and group A , using $P(Y\# = Y|X)$ instead of $P(Y\# = Y|X,A)$
- **Pros:** Guaranteed to not be making a judgement on protected attribute

What is the main limitation of fairness through unawareness?



Fairness through unawareness

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What is the main limitation of fairness through unawareness?

Hidden proxies allow the model to indirectly use protected information.



Fairness through unawareness

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 - Predict risk Y from features X and group A , using $P(Y\# = Y|X)$ instead of $P(Y\# = Y|X,A)$
- **Pros:** Guaranteed to not be making a judgement on protected attribute
- **Cons:** Other proxies may still be included in a “race-blind” setting



Group Fairness

- Idea: Require prediction rate be the same across protected groups
- E.g. “20% of the resources should go to the group that has 20% of population”
- Predict risk Y from features X and group A such that $P(Y\# = 1|A = 1) = P(Y\# = 1|A = 0)$
- Pros: Literally treats each race equally



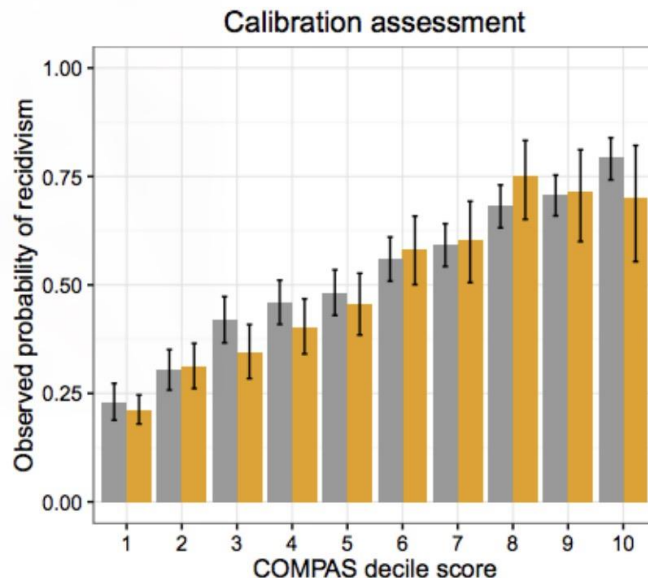
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- Pros: Literally treats each race equally
- Cons:
 - Too strong: Groups might have different base rates. Then, even a perfect classifier wouldn't qualify as “fair”
 - Too weak: Doesn't control error rate. Could be perfectly biased (correct for $A=0$ and wrong for $A=1$) and still satisfy.



Calibration

- Idea: Same positive predictive value across groups
 - Predict Y from features X and group A with score S: $P(Y = 1|S = s, A = 1) = P(Y = 1|S = s, A = 0)$
- Pros: “Equally right across groups”
- Cons: Not compatible with error rate balance



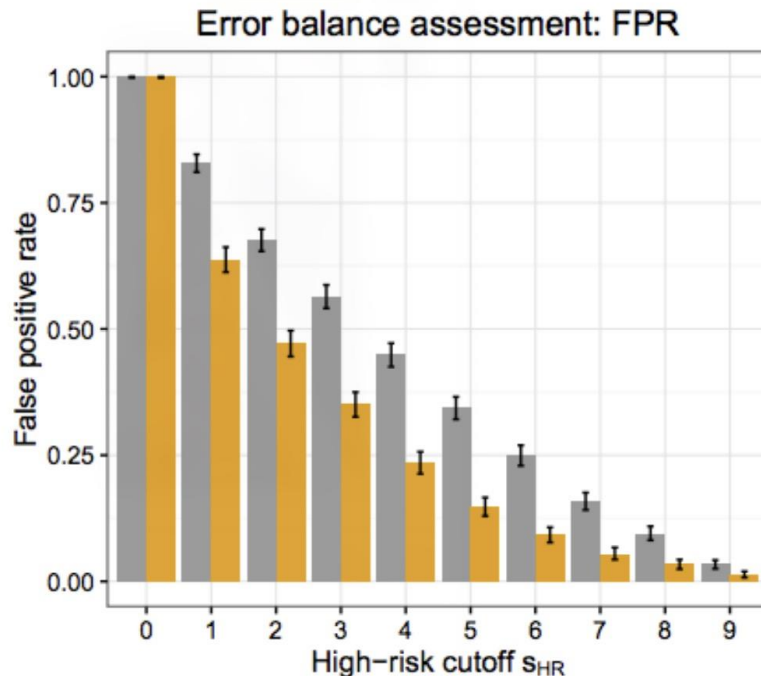
Calibration

➤ Idea: Equal false positive rates (FPR) across groups

➤ $P(Y\# = 1|Y = 0, A = 1) =$
 $P(Y\# = 1|Y = 0, A = 0)$

➤ Pros: “Equally wrong across groups”

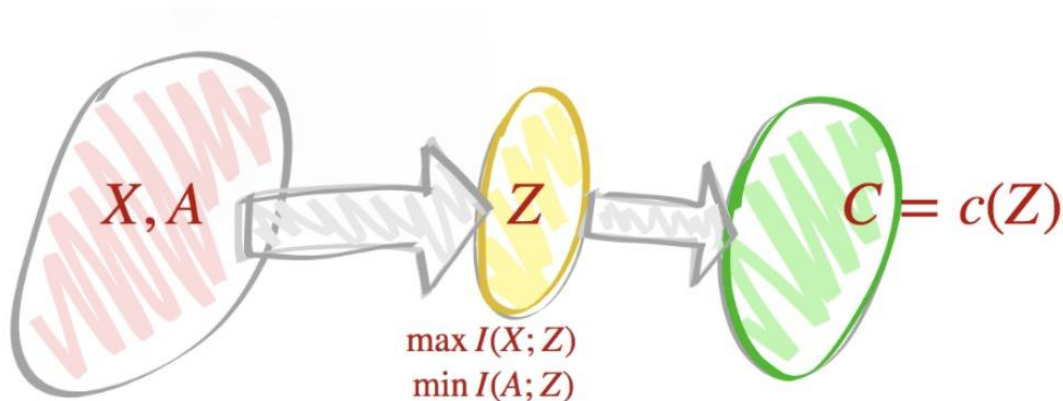
➤ Cons: Incompatible with calibration and false negative rates (FNR), could dilute with easy cases



Representational Fairness

- Idea: Learn latent representation Z to minimize group information
- Pros: Reduce information given to model but still keep important info
- Cons: Trade-off between accuracy and fairness

How does removing group information from representations help fairness?

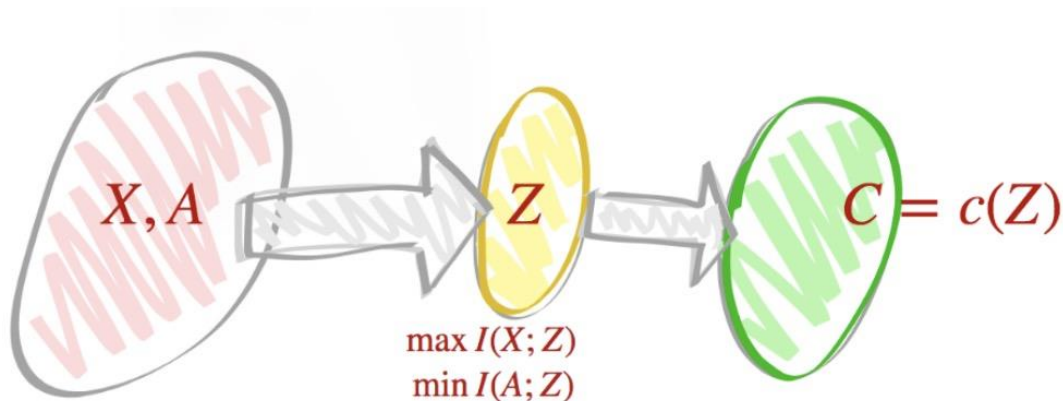


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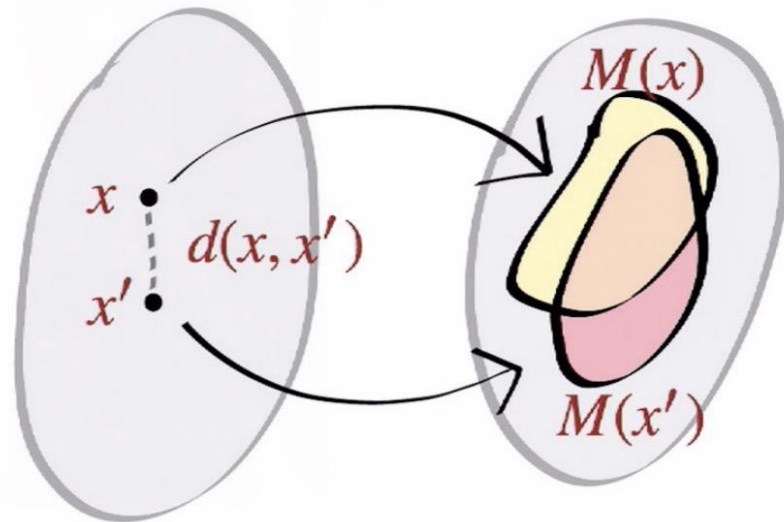
How does removing group information from representations help fairness?

Because the classifier cannot use sensitive cues if they are not encoded.



Individual fairness

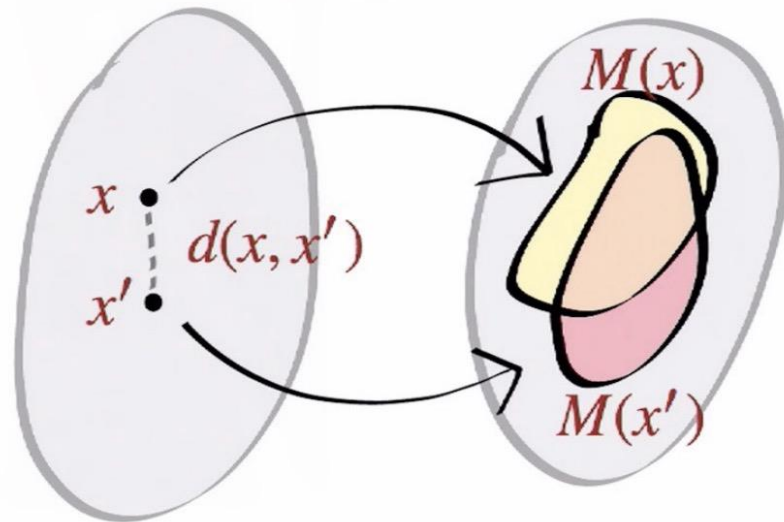
- Idea: Similar individuals should be treated similarly
- Pros: Can model heterogeneity within each group
- Cons: Notion of “similar” is hard to define mathematically, especially in high dimensions



Why is defining similarity in individual fairness challenging?

Individual fairness

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- Pros: Can model heterogeneity within each group
- Cons: Notion of “similar” is hard to define mathematically, especially in high dimensions



Why is defining similarity in individual fairness challenging?

High-dimensional data makes distance metrics ineffective and domain-specific.

Methods to improve Fairness

- **Adversarial Debiasing**

- Remove sensitive information from the learned features.
- If the representation does not contain gender/race cues, the classifier cannot use them unfairly.

We train two models simultaneously:

- **Main model:** predicts label $\hat{y}$
- **Adversary:** predicts sensitive attribute $\hat{a}$

Loss:

$$L = L_{\text{task}}(\hat{y}, y) - \lambda L_{\text{adv}}(\hat{a}, a)$$

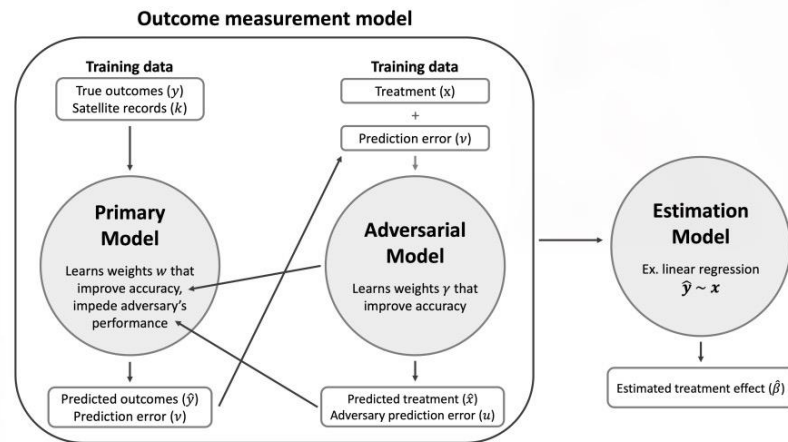


Figure 1: The model architecture of the debiasing approach

Methods to improve Fairness

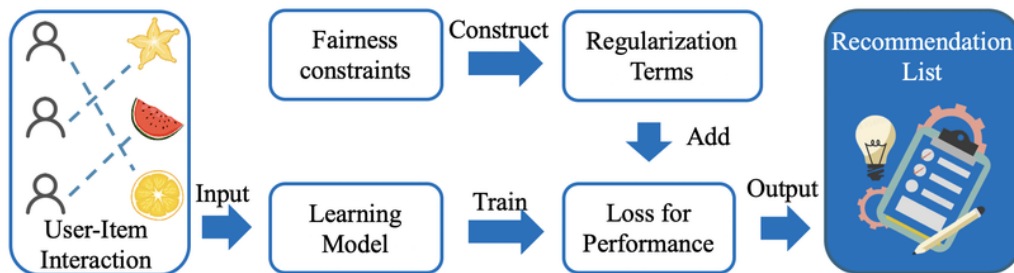
- **Fairness Regularisation (Equalised Odds, Equal Opportunity, etc.)**

Example: **Equalized Odds** requires equal TPR and FPR across groups:

$$TPR_g = TPR_{g'}, \quad FPR_g = FPR_{g'}$$

Penalty-based training:

$$L = L_{\text{task}} + \lambda \cdot [(TPR_g - TPR_{g'})^2 + (FPR_g - FPR_{g'})^2]$$



Methods to improve Fairness

- **Data Balancing + Fair Augmentations**

(a) Balancing

- Oversample minority groups
- Undersample majority groups
- Group-balanced sampling (per-batch)

Why might oversampling harm model performance?

(b) Fair Augmentations

Apply augmentations equitably across groups to reduce representation biases:

- balanced flips/crops
- lighting/pose normalization across faces
- style-transfer to equalise distribution across groups

Methods to improve Fairness

- **Data Balancing + Fair Augmentations**

(a) Balancing

- Oversample minority groups
- Undersample majority groups
- Group-balanced sampling (per-batch)

Why might oversampling harm model performance?

It increases the chance of memorising repeated examples (overfitting).

(b) Fair Augmentations

Apply augmentations equitably across groups to reduce representation biases:

- balanced flips/crops
- lighting/pose normalization across faces
- style-transfer to equalise distribution across groups

Methods to improve Fairness

- **Data Balancing + Fair Augmentations**

Pros

- Simple and highly effective.
- Does not require modifying model architecture.
- Reduces representation bias early in pipeline.

Cons

- Oversampling can cause overfitting.
- Augmentations must be carefully designed to avoid introducing new biases.

Methods to improve Fairness

- **Post-hoc Thresholding / Equalized Odds**

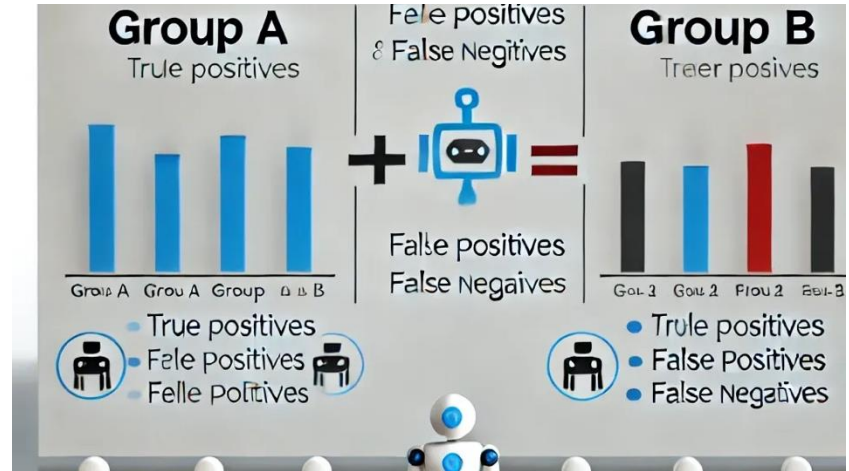
1. Compute ROC curves for each group.
2. Optimise for fairness constraints:
 - equal TPR? → Equal Opportunity
 - equal TPR + FPR? → Equalized Odds
3. Choose thresholds or flipping probabilities.

Pros

- Fast (does not retrain model).
- Achieves fairness exactly under the chosen metric.
- Popular for real-world decision-making systems.

Cons

- Predictions differ by group → may be controversial.
- Does not improve representations, only outputs.



Why Interpretability?

Interpretability

The current state of deep learning



Interpretability

And it uses



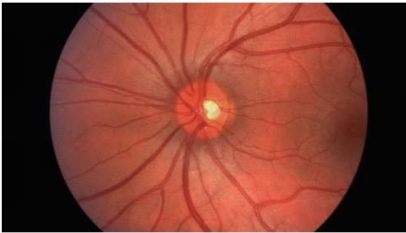
<https://www.tesla.com/videos/autopilot-self-driving-hardware-neighborhood-long>



NYPPost



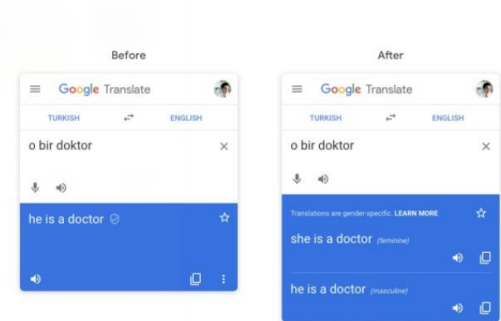
MIT Technology Review



DeepMind



DeepMind



Interpretability

What does interpretation look like?

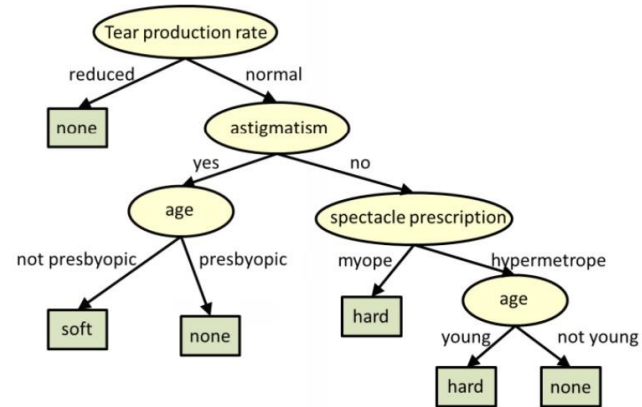
- In pre-deep learning models, some models are considered “interpretable”

Diagram illustrating the components of a linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

Labels and components:

- Dependent Variable:** Y_i
- Population Y intercept:** β_0
- Population Slope Coefficient:** β_1
- Independent Variable:** X_i
- Random Error term:** ϵ_i
- Linear component:** $\beta_0 + \beta_1 X_i$
- Random Error component:** ϵ_i



Why are deep models harder to interpret than linear models?

Interpretability

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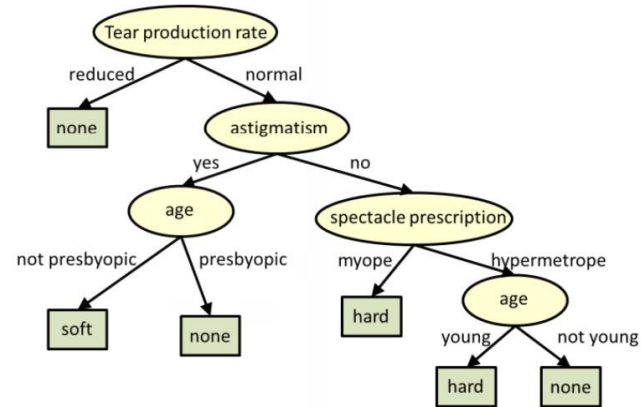
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- Random Error component:** ϵ_i



Why are deep models harder to interpret than linear models?

Because decisions depend on millions of nonlinear parameters.

Interpretability

What does interpretation look like?

- Heatmap Visualisation



Figure 3. Attribution for Diabetic Retinopathy grade prediction from a retinal fundus image. The original image is shown on the left, and the attributions (overlayed on the original image in grayscale) is shown on the right. On the original image we annotate lesions visible to a human, and confirm that the attributions indeed point to them.

[Sundarajan 2017]

in a clinical trial mainly involving patients over qqq with coronary heart disease , ramipril reduced mortality while vitamin e had no preventive effect .

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Table 2: Gate activations for each aspect in a PICC abstract. Note that because gates are calculated at the final convolution layer, activations are not in exact 1-1 correspondence with words.

[Jain 2018]

Global vs Local Interpretability

Do we explain individual prediction?

Do we explain entire model?

Example –

Heatmaps
Rationales

Example –

Prototypes
Linear Regression
Decision Trees

Which is more difficult: global or local interpretability?

Global vs Local Interpretability

Do we explain individual prediction?

Example –

Heatmaps
Rationales

Do we explain entire model?

Example –

Prototypes
Linear Regression
Decision Trees

Which is more difficult: global or local interpretability?

Global, because the full model is highly complex.

Interpretability

- Saliency Based Methods

Heatmap based visualisation
Need differentiable model in most cases
Normally involve gradient



What is the core idea behind saliency maps?

Interpretability

- Saliency Based Methods

Heatmap based visualisation
Need differentiable model in most cases
Normally involve gradient

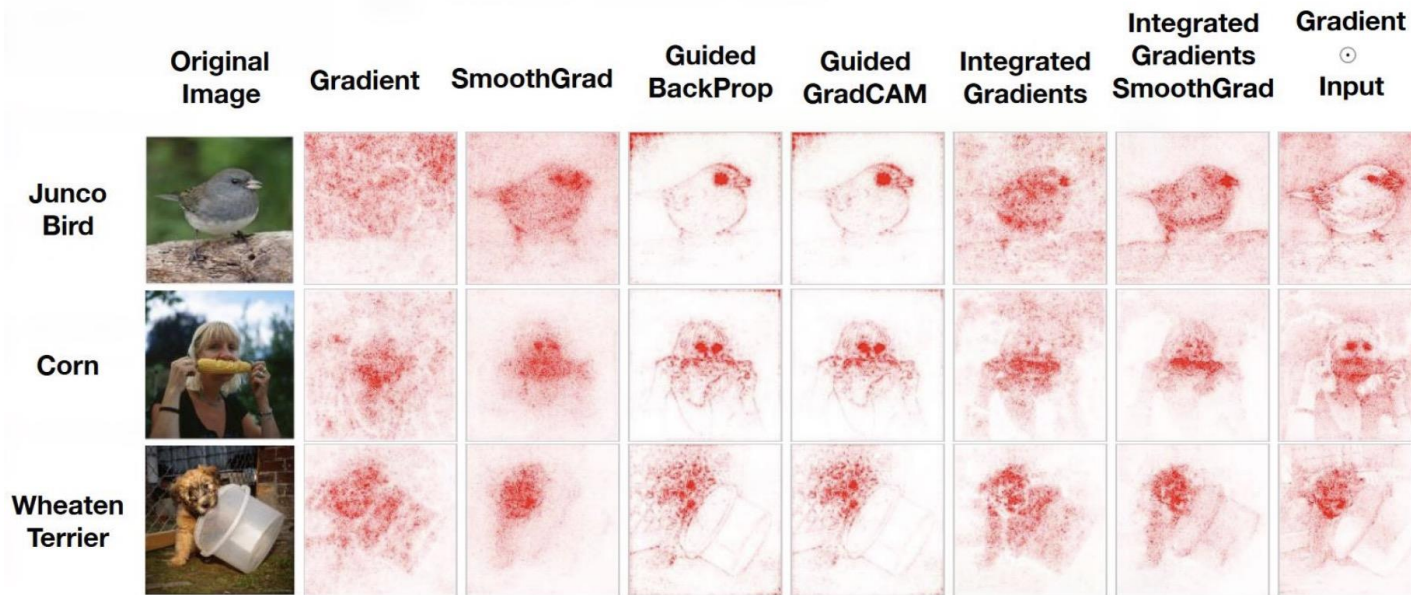


What is the core idea behind saliency maps?

Identify how changes in input affect the model output.

Interpretability

- Saliency Based Methods



[Adebayo et al 2018]

Interpretability

- Saliency Example - Gradients

$$f(x): R^d \rightarrow R$$

$$E(f)(x) = \frac{df(x)}{dx}$$

Q: How do we take gradient with respect to words?

Interpretability

- Saliency Example - Gradients

$$f(x): R^d \rightarrow R$$

$$E(f)(x) = \frac{df(x)}{dx}$$

Q: How do we take gradient with respect to words?

A: Take gradient with respect to embedding of the word.

Interpretability

- Saliency Example – Leave-one-out

$$f(x): R^d \rightarrow R$$

$$E(f)(x)_i = f(x) - f(x \setminus i)$$

How to remove ?

Interpretability

- Saliency Example – Leave-one-out

$$f(x): R^d \rightarrow R$$

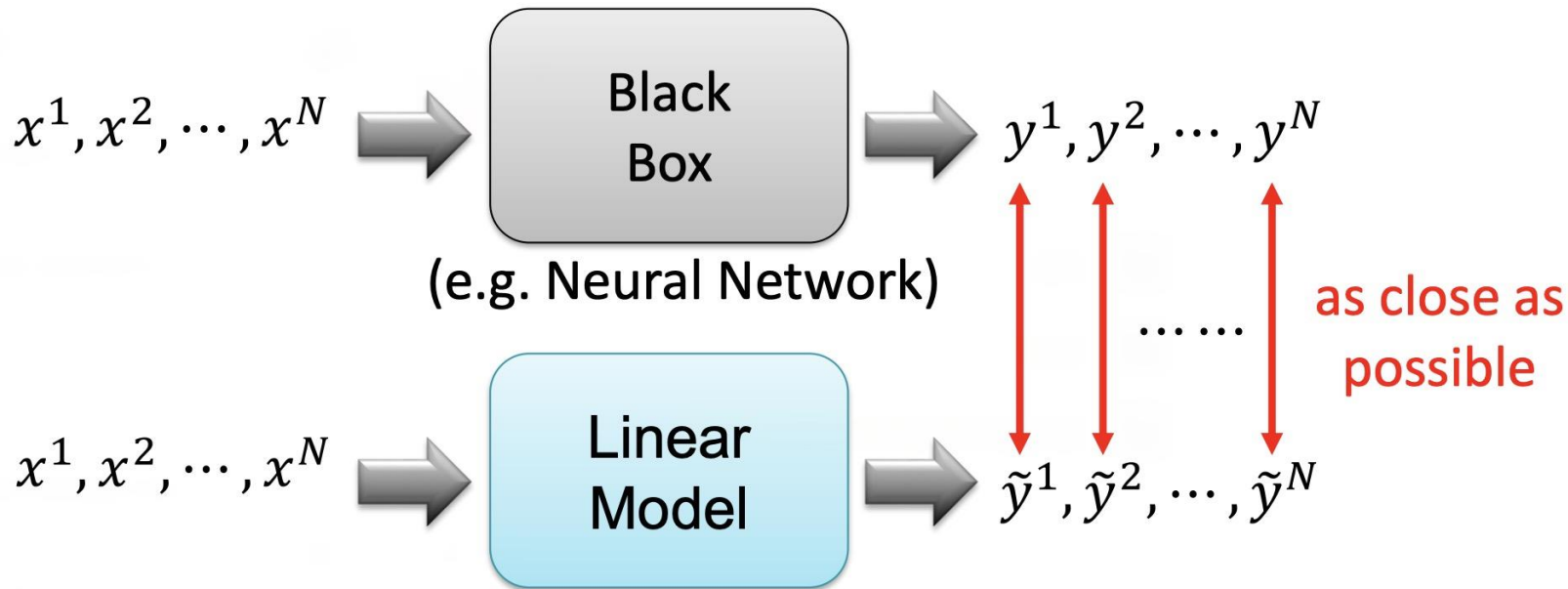
$$E(f)(x)_i = f(x) - f(x \setminus i)$$

How to remove ?

1. Zero out pixels in image
2. Remove word from the text

Interpretability

- LIME – locally interpretable model agnostic



Can't do it globally of course, but locally ? Main Idea behind LIME

Interpretability

- Intuition behind LIME

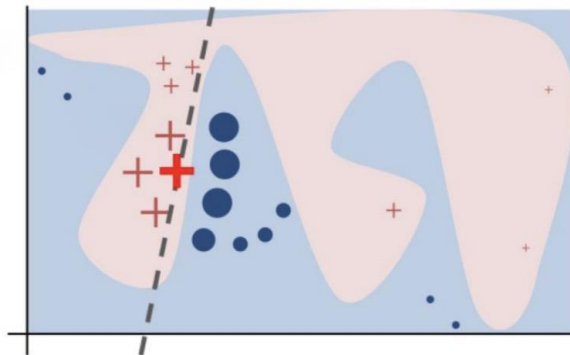
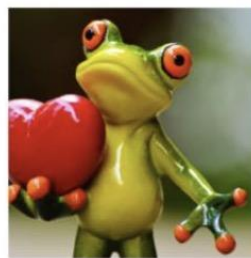


Figure 3: Toy example to present intuition for LIME. The black-box model's complex decision function f (unknown to LIME) is represented by the blue/pink background, which cannot be approximated well by a linear model. The bold red cross is the instance being explained. LIME samples instances, gets predictions using f , and weighs them by the proximity to the instance being explained (represented here by size). The dashed line is the learned explanation that is locally (but not globally) faithful.

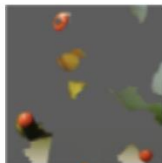
[Ribeiro et al 2016]

Interpretability

- LIME — Image



1. Given a data point you want to explain
2. Sample at the nearby - Each image is represented as a set of superpixels (segments).



Randomly delete some segments.

Black

Black

Black

0.85

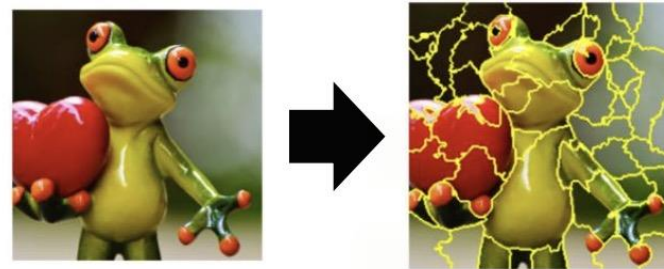
0.52

0.01

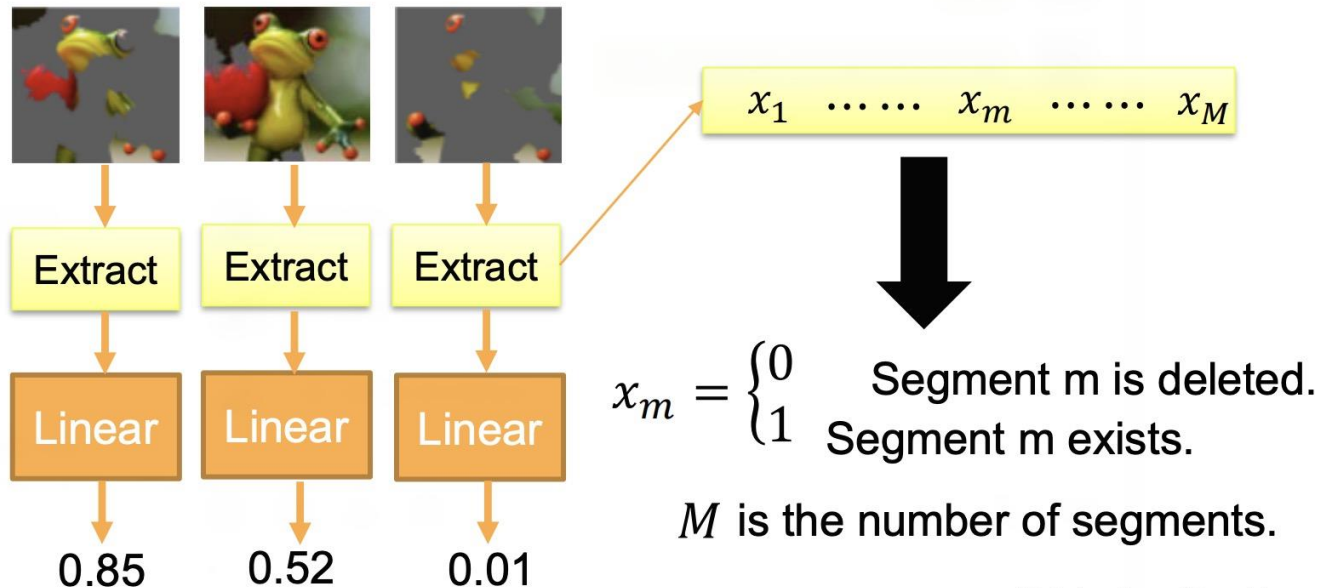
Compute the probability of “frog” by black box

Interpretability

- LIME — Image

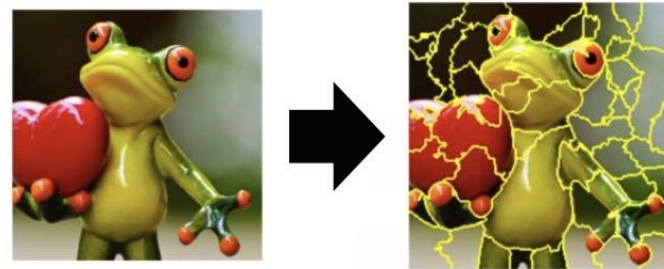


3. Fit with linear (or interpretable) model



Interpretability

- LIME — Image



4. Interpret the model you learned



Extract

Linear

0.85

$$y = w_1x_1 + \dots + w_mx_m + \dots + w_Mx_M$$

$$x_m = \begin{cases} 0 & \text{Segment } m \text{ is deleted.} \\ 1 & \text{Segment } m \text{ exists.} \end{cases}$$

M is the number of segments.

If $w_m \approx 0$ ➡ segment m is not related to “frog”

If w_m is positive ➡ segment m indicates the image is “frog”

If w_m is negative ➡ segment m indicates the image is not “frog”

Interpretability

Why can LIME never be a global explanation method?

- The Math behind LIME

Algorithm 1 Sparse Linear Explanations using LIME

Require: Classifier f , Number of samples N

Require: Instance x , and its interpretable version x'

Require: Similarity kernel π_x , Length of explanation K

$\mathcal{Z} \leftarrow \{\}$

for $i \in \{1, 2, 3, \dots, N\}$ **do**

$z'_i \leftarrow \text{sample_around}(x')$

$\mathcal{Z} \leftarrow \mathcal{Z} \cup \langle z'_i, f(z_i), \pi_x(z_i) \rangle$

end for

Match interpretable
model to black box

Control
complexity of the
model

$$\xi(x) = \operatorname{argmin}_{g \in G} \mathcal{L}(f, g, \pi_x) + \Omega(g)$$

$$\mathcal{L}(f, g, \pi_x) = \sum_{z, z' \in \mathcal{Z}} \pi_x(z) (f(z) - g(z'))^2$$

Interpretability

- The Math behind LIME

Why can LIME never be a global explanation method?

Because it approximates the model *locally*; its linear model cannot capture global nonlinearities.

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Interpretability

- SHAP

Lloyd Shapley

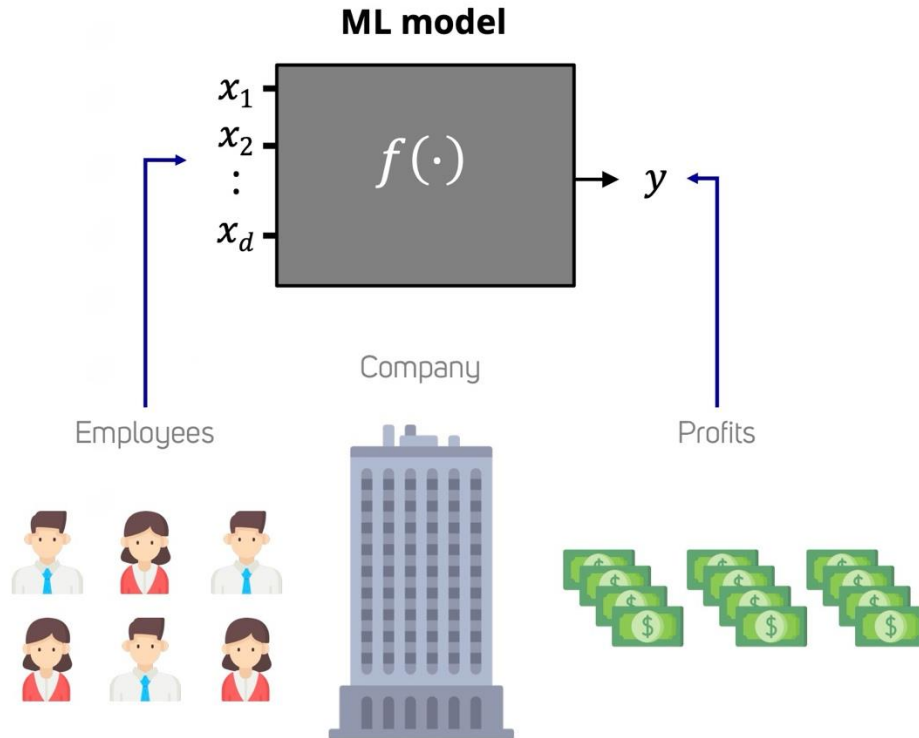
Won 2012 Nobel Memorial Prize in economics



Interpretability

- SHAP

Contribution of each input



Interpretability

- SHAP Shapley value to estimate contribution of each input

$$\phi_i(v) = \sum_{S \subseteq D \setminus i} \frac{|S|! (d - 1 - |S|)!}{d!} [v(S \cup \{i\}) - v(S)]$$

↑
Weighted
average across
 $S \subseteq D \setminus i$

↑
Contribution from
adding player i

Why are Shapley values considered fair?

Interpretability

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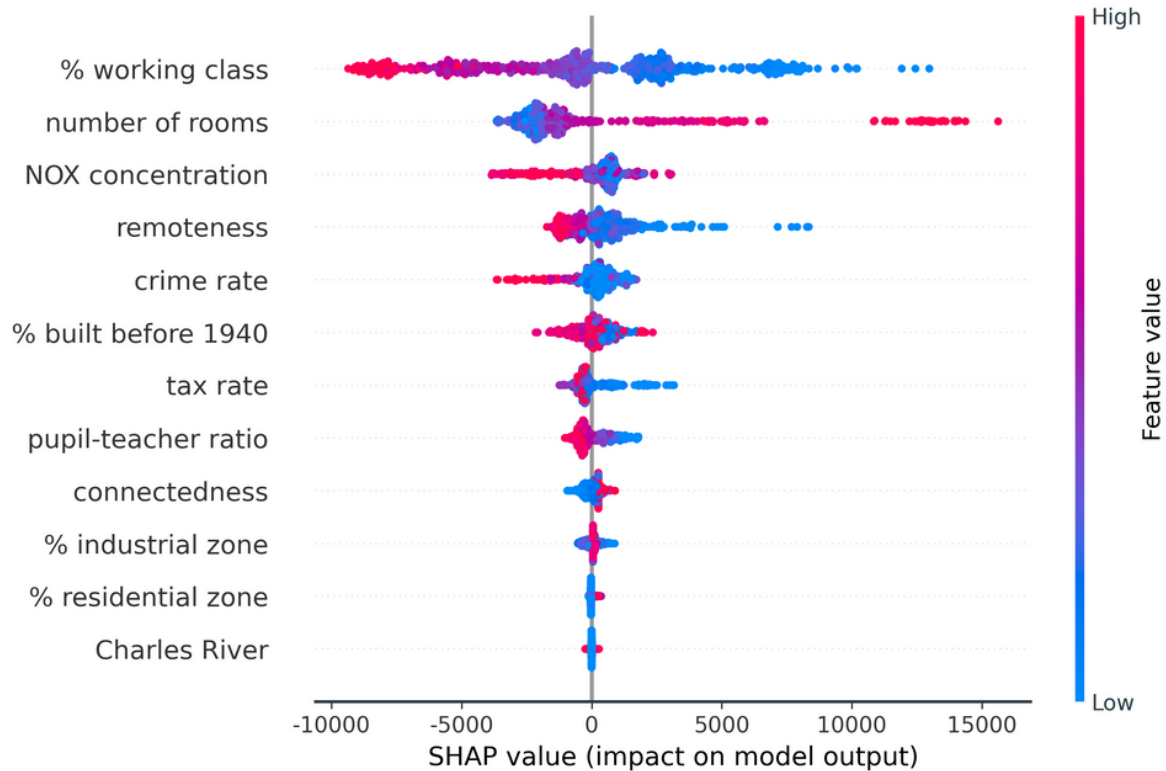
↑
Contribution from
adding player i

Why are Shapley values considered fair?

Because they average contributions across all possible coalitions of features.

Interpretability

- SHAP



Interpretability

- Global Interpretability – Information Bottleneck

We phrase the *learning problem* as a *mapping* problem:

$$X \rightarrow Y$$

- X,Y are two (potentially high-dimensional) variables
- X may be images/genomes/videoframes,
Y may be categorical labels/expression levels/reward signals.

Interpretability

- Global Interpretability – Information Bottleneck

Further, let us assume that we may solve the *learning problem* using *intermediate representations*:

$$X \rightarrow Z \rightarrow Y$$

- Intermediate representation inspired by real(!) neural networks
- Z encodes efficiently X (*compression*)
Z eases mapping onto Y (*relevance*)

Interpretability

- Global Interpretability – Information Bottleneck

How can we compute *optimal intermediate representations* Z ?

We want to Z to contain all and only the information relevant to Y :

$$\min \underbrace{I[X; Z]}_{\text{compression}} \quad \max \underbrace{I[Z; Y]}_{\text{relevance}}$$

- We maximize the compression by *minimizing the mutual information between X and Z*
- We maximize the relevance by *maximizing the mutual information between Z and Y* (Infomax principle [4])
- (Information theory $\rightarrow$ *rate-distortion theory*)
- (Statistics $\rightarrow$ *sufficient statistics*)

Interpretability

- Global Interpretability – Information Bottleneck

We can re-express our objective as a *single optimization problem*:

$$\arg \min_Z I[X; Z] - \beta I[Z; Y]$$

- Optimization is wrt Z (it may be a parametric representation)
- β is a Lagrangian and trades off compression and relevance
- (This has an analytic solution using *Blahut-Arimoto algorithm* [14])
- (Practically, though, estimating mutual information is hard)

Interpretability

- Global Interpretability – Information Bottleneck

Can we explain learning in deep neural networks using IB?

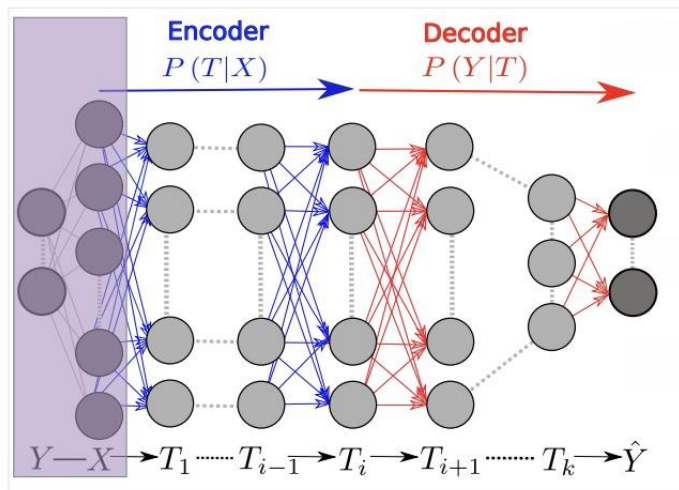


Image from [10]

- Every layer of the network computes an intermediate representation Z_i

Interpretability

- Global Interpretability – Information Bottleneck

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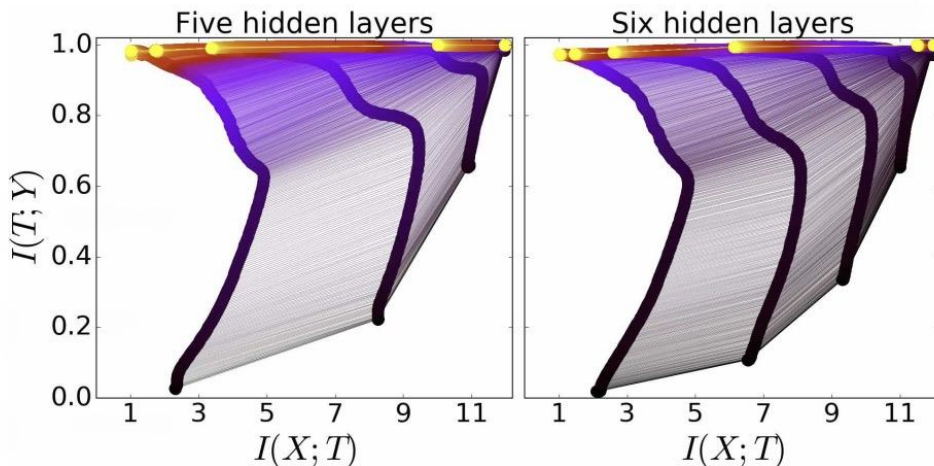


Image from [10]

- Trajectory in the information plane agrees with IB theory
- (Two different learning phases may be identified)
- (There are some criticisms of this analysis)

Next: Revision